

Structural reasoning — or surface-level sensitivity?

Evaluating Robustness of Reasoning Models on Parameterized Logical Problems

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01 INTRODUCTION

Reasoning LLMs ace combinatorial tasks — until they don't.

On planning, puzzles and SAT, large reasoning models (LRMs) outperform standard LLMs, then degrade abruptly past a complexity threshold.

Two competing readings

- **Optimistic.** LRMs approximate solver-like search through self-correction and candidate selection — accuracy should stay stable under semantics-preserving shifts and move only with structural difficulty.
- **Skeptical.** Gains come from replaying memorized trace templates, not from systematic reasoning.

Existing benchmarks can't decide.

Logic and SAT-in-natural-language benchmarks mostly vary surface wording, not structure. We need a setting where difficulty is tuned along interpretable structural axes while satisfiability is preserved.

Our contribution.

A parameterised 2-SAT diagnostic that pulls these readings apart — and shows that today's reasoning LLMs sit much closer to the skeptical one than aggregate accuracy suggests.

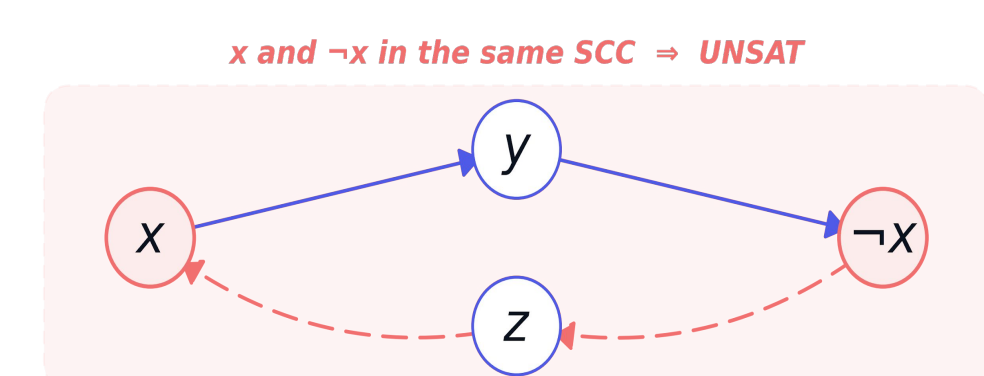
02 Why 2-SAT

Polynomial-time, yet structurally rich.

2-SAT is solvable in linear time — so failure cannot be blamed on raw computational hardness. It reveals a reasoning gap, not a search gap.

Exact graph characterisation.

Build the implication graph: each clause $(\ell_1 \vee \ell_2)$ contributes the edges $\neg\ell_1 \rightarrow \ell_2$ and $\neg\ell_2 \rightarrow \ell_1$.



Tunable along interpretable axes.

Cycle length, backbone fraction, free-variable count, bridge position, clause repetition — each axis is a surgical probe for one reasoning competency.

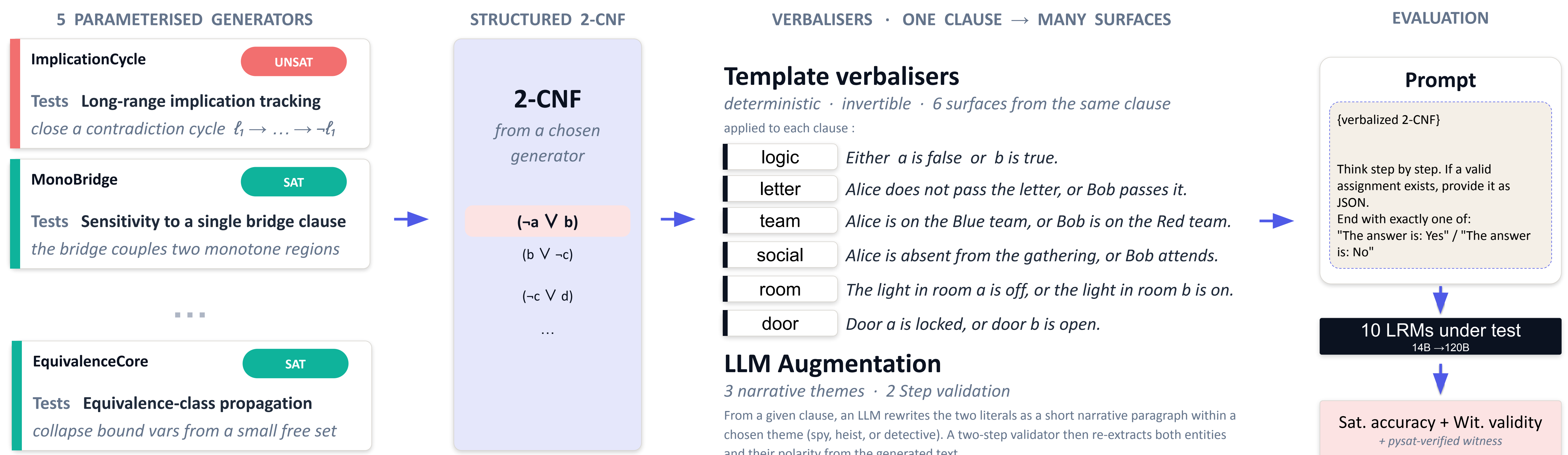
Witnesses are cheap.

Every SAT prediction is verified deterministically against the CNF. We score decision accuracy and witness validity as two separate metrics — they often disagree.

The simplicity is the test.

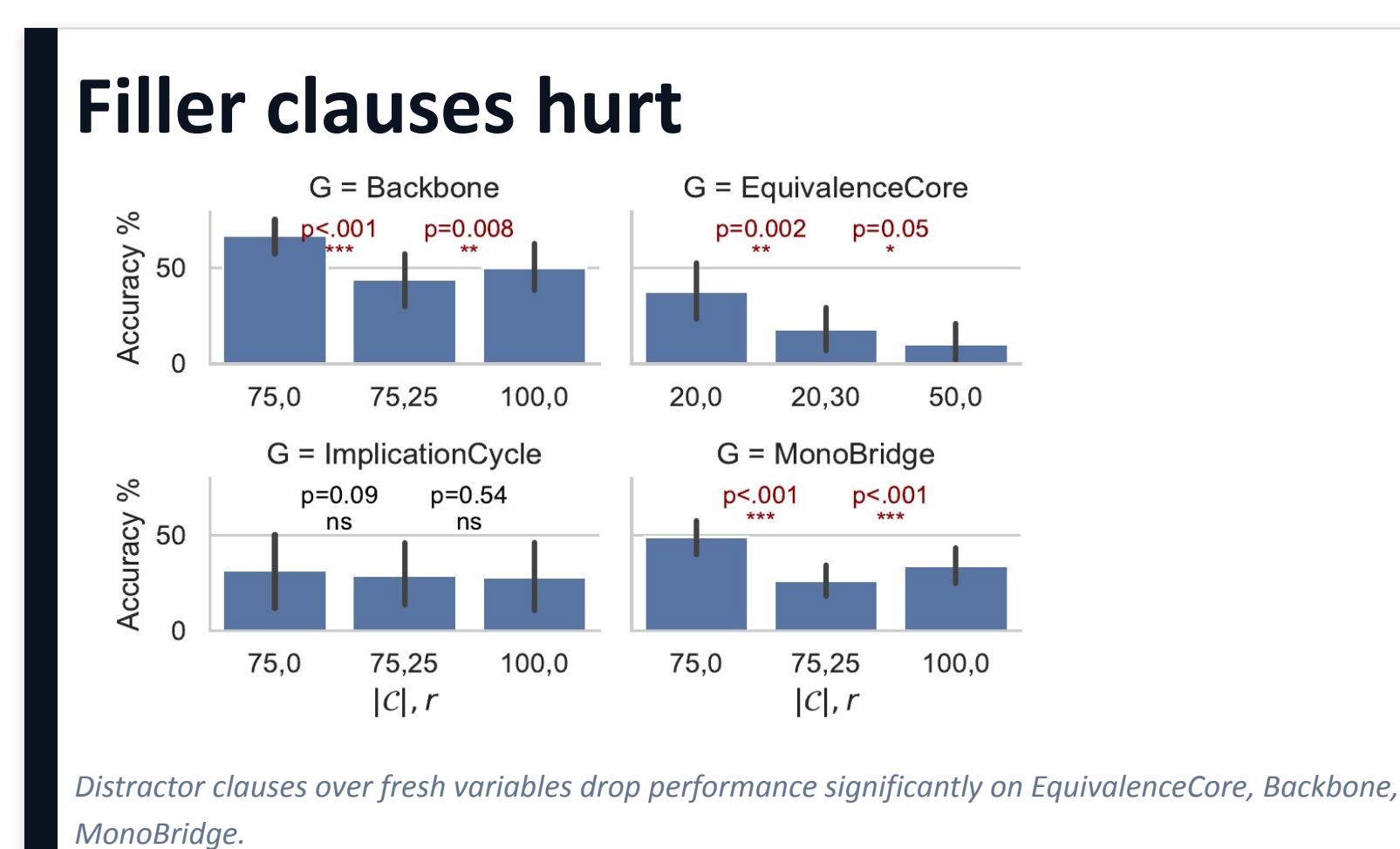
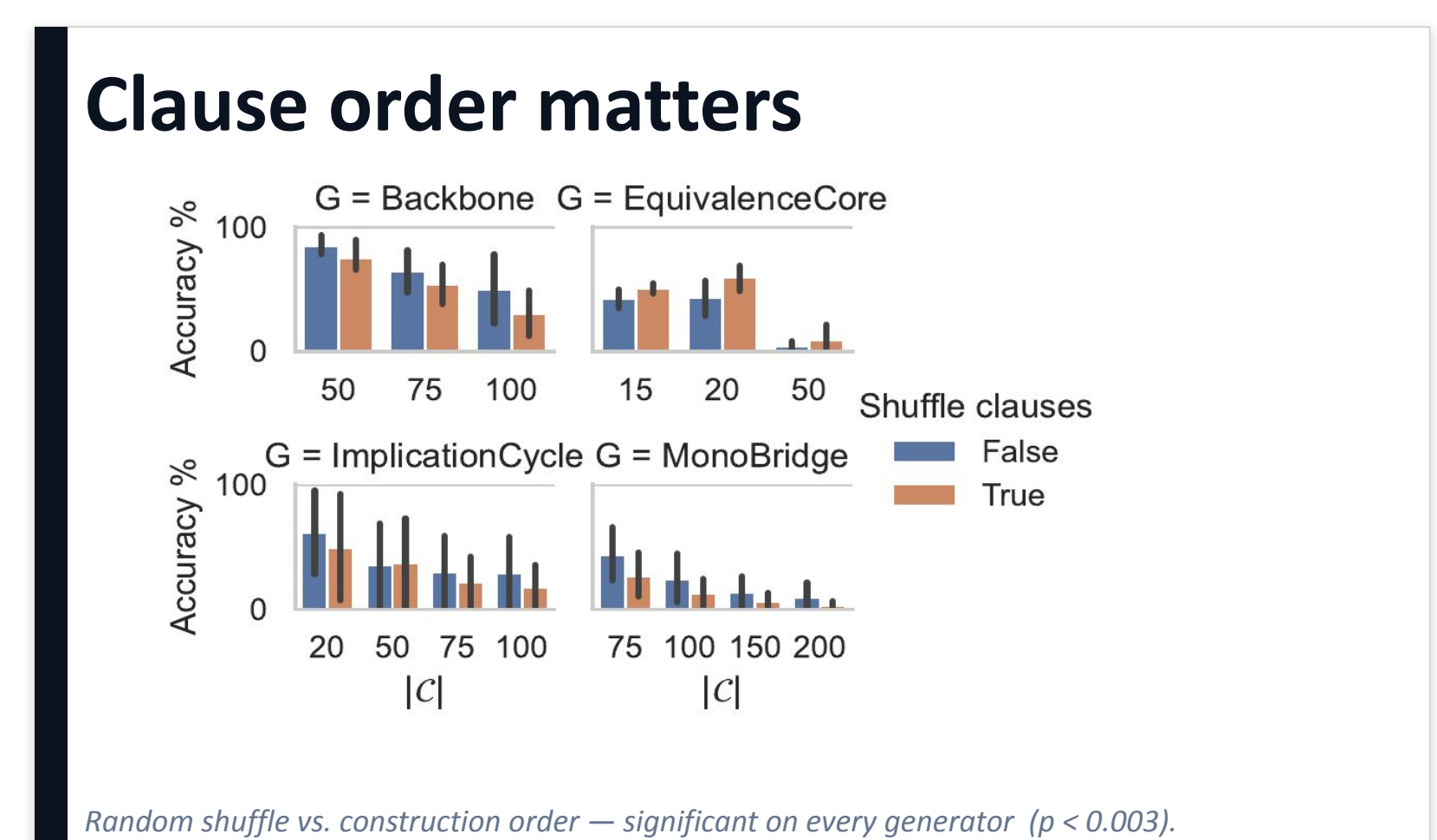
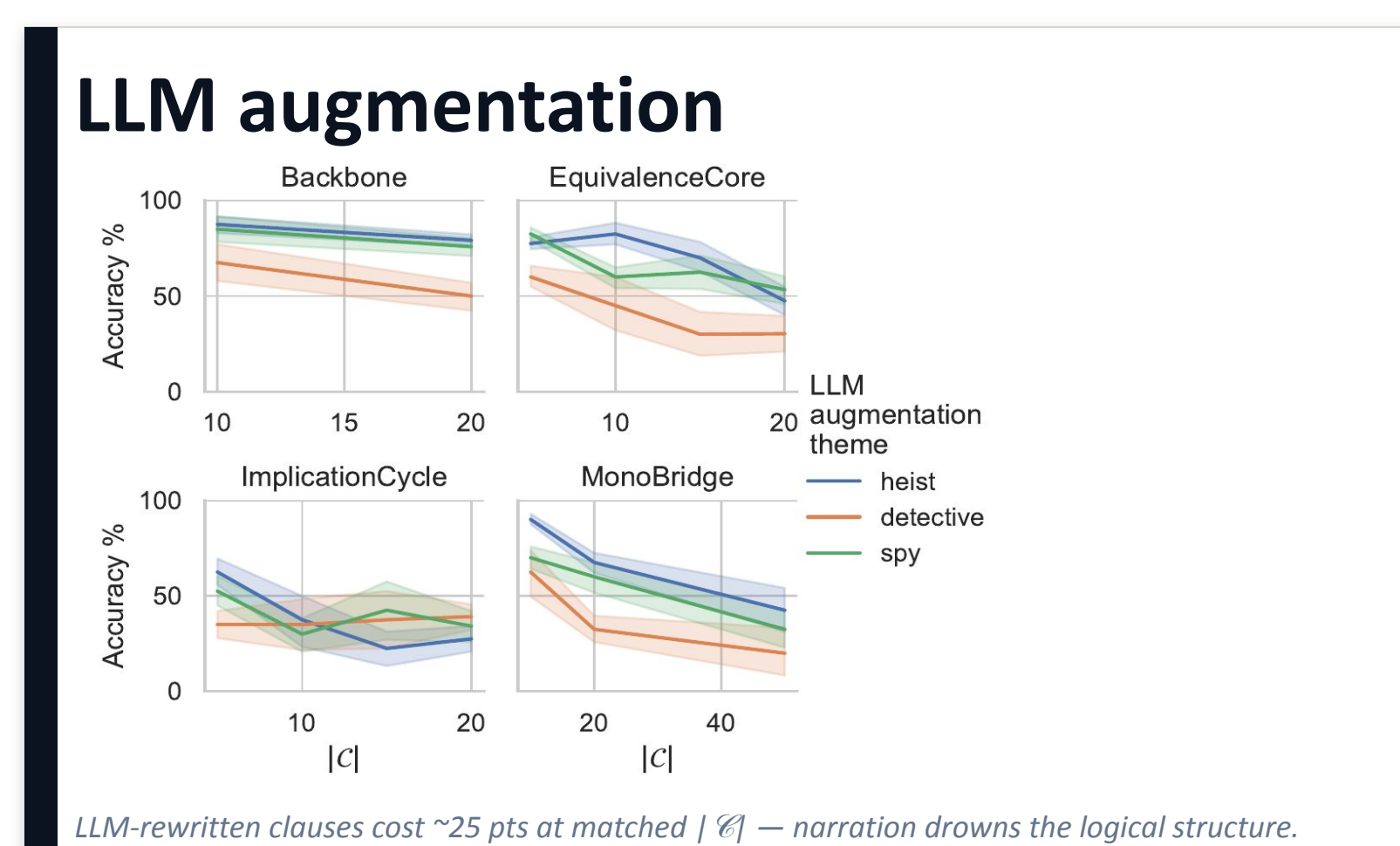
That 2-SAT is easy for SAT solvers is exactly the point — it isolates structural reasoning from search hardness. Moving to 3-SAT would conflate the two.

03 PIPELINE



04 RESULTS

	Model c	Llama-3.3 70B-Instruct		OLMo-3 32B		QwQ 32B		Qwen3-Next 80B		GPT-OSS 120B	
		Sat.	Wit.	Sat.	Wit.	Sat.	Wit.	Sat.	Wit.	Sat.	Wit.
Implication Cycle	5	0.0	—	98.3	—	98.3	—	100.0	—	91.7	—
	10	1.7	—	95.0	—	96.7	—	98.3	—	95.0	—
	15	15.0	—	96.7	—	68.3	—	96.7	—	93.3	—
	20	18.3	—	88.3	—	65.0	—	96.7	—	93.3	—
	50	11.7	—	65.0	—	40.0	—	81.7	—	56.7	—
	75	13.3	—	50.0	—	48.3	—	73.3	—	45.0	—
Equivalence Core	100	20.0	—	51.7	—	45.0	—	76.7	—	40.0	—
	10	100.0	1.7	100.0	93.3	100.0	86.7	100.0	91.7	100.0	83.3
	15	98.3	0.0	100.0	55.0	96.7	45.0	100.0	53.3	100.0	46.7
	20	96.7	0.0	91.7	56.7	86.7	43.3	100.0	63.3	100.0	50.0
	50	93.3	0.0	83.3	1.7	83.3	1.7	98.3	8.3	100.0	8.3
	20	100.0	63.3	100.0	96.7	100.0	98.3	100.0	96.7	100.0	96.7
Backbone	50	98.3	71.7	98.3	83.3	98.3	76.7	100.0	76.7	100.0	98.3
	75	96.7	51.7	90.0	36.7	96.7	65.0	100.0	75.0	98.3	88.3
	100	100.0	50.0	83.3	15.0	93.3	50.0	98.3	73.3	100.0	83.3
	10	100.0	55.0	100.0	98.3	100.0	90.0	100.0	100.0	100.0	100.0
	20	100.0	36.7	100.0	96.7	98.3	86.7	100.0	100.0	100.0	100.0
	50	98.3	13.3	96.7	66.7	96.7	58.3	100.0	61.7	100.0	83.3
Mono Bridge	75	100.0	13.3	95.0	18.3	68.3	28.3	93.3	60.0	100.0	68.3
	100	100.0	1.7	88.3	1.7	83.3	13.3	91.7	23.3	100.0	53.3
	150	96.7	3.3	80.0	1.7	78.3	1.7	93.3	21.7	90.0	31.7
	200	98.3	0.0	66.7	0.0	83.3	0.0	93.3	16.7	73.3	23.3



Key takeaways

- **Sat. ≠ Wit.** Decision accuracy hides weak witness construction
- **Long contradiction chains break every model.** Tracking $x \rightarrow \dots \rightarrow \neg x$ over 50+ implications is where all 7 LRMs fail.
- **-25 pt under LLM narrative** the LLM verbaliser drowns the logical structure.
- **Order matters** clause shuffle hurts on all generators
- **Repetition is invisible** duplicating clauses $(\Psi \wedge p(\Psi))$ costs as much as adding new ones.

